2 The Counting Curve

2.1 Equipment Setup

2.1.1 Required Equipment

- radiation sources
- proportional counter
- oscilloscope
- pre-amplifier
- amplifier
- single channel analyzer
- counter/timer
- multi-channel analyzer

2.2 Optimizing LLD, Gain, and Shaping Time

It is important that you select a suitable gain and shaping time for your device before you begin the counting curve experiments (Section 2.3).

- Connect the proportional counter to the preamp.
- Connect the preamp to HV, and route the output to amplifier.
- Split the amplifier output to the scope.
- Connect the output of the amplifier to the MCA.

2.2.1 Getting familiar with the gas

- Ensure that your system is properly connected and that there are no sources in your detector. Use a HV of +800 V for this section.
- Open Maestro on your computer and set the live time to have no limit
- Place an Am-241 source in the chamber.
- Ask your GSI to increase the flow rate to maximum.
- Begin collecting data with Maestro.

You should now see the upper edge of the spectrum in Maestro increasing over time. This is due to the P-10 gas flushing through the system to ensure proper multiplication. If you cannot clearly see the peak after a minute with the gas turned up, ask your GSI for help if necessary.

2.2.2 Determine the ideal shaping time

You will use a single shaping time throughout this experiment. This section will help you select the ideal shaping time at the beginning.

- Adjust the coarse gain and fine gain such that the alpha peak occurs around the center of the spectrum.
- For each shaping time option on your amplifier, perform the following steps:
  - Record a ~1 minute run in Maestro
  - Set an ROI around the peak (drag your mouse over the alpha peak and select mark ROI). Be sure that the ROI includes the entire peak, but do not extend it too far to either side.
  - Double click the ROI and record the centroid channel and gross area of the alpha peak.

Now pick the best shaping time based on what your learned in this course to date. You will want to select the shortest shaping time that does not exhibit a significant ballistic deficit. Confirm